

Zebin Yang

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ABOUT ME

I am a PH.D. student at SEC Lab at Peking University, advised by Prof. Meng Li. My current research interest lies in **Embodied AI**, **LLM Inference Acceleration**, **Model Compression**, and other emerging technologies. I received my bachelor's degree from Peking University under the supervision of Prof. Meng Li.

EDUCATION

Peking University, Beijing, China

Sept. 2024 - Now

Ph.D., School of Integrated Circuits

Peking University, Beijing, China

Sept. 2020 - Jun. 2024

B.S., School of Electronic Engineering and Computer Science (**with Honors**)

RESEARCH INTERNSHIPS & COLLABORATIONS

PrimeBot Research Institute, China

Mar. 2026 - Jun. 2026

Advised by Jiafeng Xu, Hao Dong

Edge deployment and inference optimization for VLA models. Research output: **JetsonPI** [1].

Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), Shenzhen, China Sept. 2024 - Sept. 2025

Advised by Bo Yu, Shaoshan Liu

Long horizon robot navigation and planning optimization. Research outputs: **EfficientNav** [4], **KEEP** [2].

RESEARCH INTERESTS

The listed references mean the relevant works I have participated in.

- **Efficient Embodied AI System**: Investigate efficient VLA inference and efficient embodied planning.
 - **Asynchronous VLA inference**: decouple VLA inference from action execution to enable continuous, real-time robot control, while addressing perception-execution misalignment and reducing reaction time through foresight-aligned correction and scheduling optimization [1].
 - **VLA inference engine**: develop Jetson-PI-Edge, an efficient llama.cpp-based inference engine for real-time VLA deployment on edge platforms, including desktop GPUs, CPUs, NPUs, and NVIDIA Jetson Orin/Thor. On Jetson Orin, Jetson-PI-Edge reduces $\pi_{0.5}$ latency by up to $3.44\times$ over PyTorch [1].
 - **VLA acceleration**: develop a layer-skipping framework that skips redundant computation according to action importance, achieving up to $3.75\times$ inference speedup and $85.7\times$ training-cost reduction while maintaining competitive manipulation performance [3].
 - **Memory management for embodied planning**: investigate efficient memory management methods for long-term embodied tasks such as navigation and household works, reducing planning latency by KV cache management, built on vLLM framework [2][4].
- **Model Compression & Inference Acceleration**: Design end-to-end optimization pipelines to improve inference efficiency across LLMs and linear transformers.
 - **KV cache management for LLM inference**: investigate efficient KV cache optimizations, including re-computation, position-independent caching, block attention [2][4].
 - **Quantization**: Optimize weight and activation quantization for fast inference; explore channel-adaptive precision allocation to balance model accuracy and computational overhead [5][6][8].
 - **Knowledge distillation (KD)**: employ logits-based distillation and feature-based distillation for model compression [5].
 - **Neural architecture search (NAS)**: automatically prunes redundant neurons for model compression [5].

PUBLICATIONS

** indicates equal contribution.*

The publication list is available through Google Scholar: please click here.

[1] **Jetson-PI: Towards Onboard Real-Time Robot Control via Foresight-Aligned Asynchronous Inference.**

Zebin Yang, Qi Wang, Yunhe Wang, Xiurui Guo, Bo Yu, Shaoshan Liu, Jiafeng Xu, Hao Dong, Meng Li
CoRL'26, GitHub 1, GitHub 2

[2] **KEEP: A KV-Cache-Centric Memory Management System for Efficient Embodied Planning.**

Zebin Yang, Tong Xie, Baotong Lu, Bo Yu, Shaoshan Liu, Meng Li
DAC'26, CCF-A, GitHub

[3] **DySL-VLA: Efficient Vision-Language-Action Model Inference via Dynamic-Static Layer-Skipping for Robot Manipulation.**

Zebin Yang, Yijiahao Qi, Tong Xie, Bo Yu, Shaoshan Liu, Meng Li
DAC'26, CCF-A, GitHub

[4] **EfficientNav: Towards On-Device Object-Goal Navigation with Navigation Map Caching and Retrieval.**

Zebin Yang, Sunjian Zheng, Tong Xie, Tianshi Xu, Bo Yu, Fan Wang, Jie Tang, Shaoshan Liu, Meng Li
NeurIPS'25, CCF-A, GitHub

[5] **MCUBERT: Memory-Efficient BERT Inference on Commodity Microcontrollers.**

Zebin Yang*, Renze Chen*, Taiqiang Wu, Ngai Wong, Yun (Eric) Liang, Runsheng Wang, Ru Huang, Meng Li
ICCAD'24, CCF-B

[6] **FastQuery: Communication-efficient Embedding Table Query for Private LLM Inference.**

Chenqi Lin*, Tianshi Xu*, **Zebin Yang***, Runsheng Wang, Ru Huang, Meng Li
DAC'24, CCF-A

[7] **ProPD: Dynamic Token Tree Pruning and Generation for LLM Parallel Decoding.**

Shuzhang Zhong, **Zebin Yang**, Ruihao Gong, Runsheng Wang, Ru Huang, Meng Li
ICCAD'24, CCF-B

[8] **LightMamba: Efficient Mamba Acceleration on FPGA with Quantization and Hardware Co-design.**

Renjie Wei*, Linfeng Zhong*, Songqiang Xu*, **Zebin Yang**, Qingyu Guo, Yuan Wang, Runsheng Wang, Meng Li
DATE'25, CCF-B, GitHub

HONORS AND ACHIEVEMENTS

Outstanding Graduates at Peking University	Jul. 2024
Merit Student	Sep. 2022,2023
UBIQUANT scholarship	Dec. 2023
Lee Wai Wing scholarship	Dec. 2022

COMPETITION AWARDS

The 13th Chinese Mathematics Competitions (CMC)	2022
First Prize	
2022 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) (Higher Education Press Cup)	2022
Second Prize	
The 38th National Regions Physics Competition	2022
Second Prize	
2021 Asia and Pacific Mathematical Contest in Modeling (APMCM)	2021
Second Prize	